CAPSTONE PROJECT FINAL REPORT

SMART GREENHOUSE MONITORING SYSTEM

Cloud-Based IoT Monitoring with 3D Visualization

Course Code: PROJ3999

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Live Demo: https://greenhouse-pi.vercel.app/

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CHAPTER 1: ABSTRACT

The Smart Greenhouse Monitoring System is a cloud-based Internet of Things (IoT) web application designed to provide real-time monitoring, control, and visualization of greenhouse environments. The system addresses the growing need for precision agriculture by integrating hardware sensors (temperature, humidity, soil moisture, and light) with a modern web dashboard that enables remote monitoring and automated control of greenhouse operations.

The application was developed using React 18 with TypeScript for the frontend, Supabase for backend database and authentication services, and Three.js (React Three Fiber) for advanced 3D plant lifecycle simulations. The system supports 15 distinct plant species with unique growth patterns, environmental response algorithms, and a side-by-side comparison mode. IoT sensor data is ingested through Supabase Edge Functions, enabling real-time data processing, threshold-based alerting, and historical trend analysis.

Key findings demonstrate that the system successfully provides a comprehensive greenhouse management platform with live sensor monitoring, automated scheduling for irrigation, lighting, ventilation, and heating, interactive circuit diagrams, and data export capabilities. The 3D simulation accurately models plant growth influenced by temperature, humidity, light, rain, and grow light conditions, providing a valuable educational and planning tool for greenhouse operators.

The significance of this project lies in its demonstration of how modern web technologies, cloud computing, and IoT can be combined to create an accessible, cost-effective solution for precision agriculture, contributing to food security and sustainable farming practices.

CHAPTER 2: INTRODUCTION

2.1 BACKGROUND

Agriculture has been the backbone of human civilization, and greenhouses have played a crucial role in extending growing seasons, protecting crops from adverse weather, and enabling year-round food production. However, traditional greenhouse management relies heavily on manual monitoring and control, which is labor-intensive, error-prone, and inefficient. With the advent of the Internet of Things (IoT) and cloud computing, there is a significant opportunity to automate and optimize greenhouse operations.

Previous research and commercial products in smart agriculture have demonstrated the feasibility of sensor-based monitoring systems. However, many existing solutions are either proprietary and expensive, lack comprehensive visualization capabilities, or do not provide the level of integration needed for small-to-medium scale greenhouse operations. This project builds upon existing work in IoT-based agriculture while introducing novel features such as 3D plant lifecycle simulation, real-time sensor data synchronization with virtual environments, and a comparison mode for different plant species.

2.2 PURPOSE

The purpose of this project is to design and develop a comprehensive Smart Greenhouse Monitoring System that enables greenhouse operators to:

- Monitor environmental parameters (temperature, humidity, soil moisture, light) in real-time through an intuitive web dashboard.
- Control greenhouse actuators (irrigation, lighting, ventilation, heating, cooling, misting) remotely via toggle controls.
- Visualize plant growth through interactive 3D simulations that respond to real environmental conditions.
- Schedule automated tasks for irrigation, lighting, and ventilation with a calendar-based interface.
- Receive automated alerts when sensor readings exceed predefined thresholds.
- Register and manage IoT devices with status monitoring, battery level, and signal strength tracking.
- Export historical sensor data for analysis and record-keeping.

2.3 SCOPE

The scope of this project encompasses the complete development lifecycle from requirements analysis to deployment. The system is designed as a web application accessible through modern web browsers, eliminating the need for native application installation. The project scope includes:

- A responsive web dashboard with six main modules: Dashboard, Plants, 3D View, IoT Devices, Schedules, and Analytics.
- Integration with IoT sensors via RESTful API endpoints (Supabase Edge Functions) for data ingestion.
- Cloud-based data storage using PostgreSQL (via Supabase) with Row-Level Security (RLS) policies.
- User authentication with email verification for secure access control.
- 3D visualization engine supporting 15 plant species with environment-responsive growth simulation.
- A comparison mode for side-by-side analysis of different plant species under identical conditions.

Limitations include the dependency on internet connectivity for cloud communication, browser compatibility requirements for Three.js rendering, and the simulation nature of some sensor data in the demonstration environment.

CHAPTER 3: LITERATURE SURVEY

This chapter reviews existing literature and projects related to smart greenhouse monitoring systems, IoT in agriculture, and 3D visualization in environmental simulation.

3.1 IoT-Based Smart Agriculture Systems: Numerous studies have explored the application of IoT in agriculture. Kodali and Sahu (2016) presented an IoT-based weather monitoring system using Raspberry Pi that collected temperature and humidity data. Their work demonstrated the feasibility of low-cost sensor networks but lacked a comprehensive dashboard and visualization component. Similarly, Rajalakshmi and Mahalakshmi (2016) implemented an IoT-based crop-field monitoring system that used soil moisture sensors to automate irrigation, achieving a 30% reduction in water consumption.

3.2 Cloud-Based Monitoring Platforms: The integration of cloud computing with IoT has been extensively studied. Gubbi et al. (2013) provided a comprehensive review of IoT architectures and highlighted the importance of cloud backends for scalability and data processing. Amazon Web Services, Google Cloud IoT, and Microsoft Azure IoT Hub have emerged as leading platforms for IoT data management. This project utilizes Supabase, an open-source alternative to Firebase, which provides PostgreSQL database, authentication, real-time subscriptions, and edge functions in a unified platform.

3.3 Greenhouse Automation and Control: Automated greenhouse control systems have evolved significantly. Hemming et al. (2019) developed an AI-driven greenhouse climate control system that optimized energy consumption while maintaining optimal growing conditions. Their work emphasized the importance of threshold-based alerting and automated response systems, concepts that are implemented in our project through sensor threshold monitoring and automated alert generation via Edge Functions.

3.4 3D Visualization in Agriculture: Three-dimensional visualization has been increasingly used in agricultural simulation. Prusinkiewicz and Lindenmayer (1990) pioneered L-system-based plant growth modeling, which inspired procedural geometry approaches used in game development and scientific visualization. More recently, WebGL and Three.js have enabled browser-based 3D rendering without plugin requirements. This project leverages React Three Fiber, a React renderer for Three.js, to create interactive 3D plant lifecycle simulations with environment-responsive growth algorithms.

3.5 Web Technologies for Dashboard Development: Modern web frameworks have significantly improved the development of data-intensive dashboards. React, developed by Meta, provides component-based architecture and efficient virtual DOM rendering. Combined with data visualization

libraries such as Recharts and D3.js, React enables the creation of responsive, real-time dashboards. Tailwind CSS provides utility-first styling that accelerates UI development while maintaining design consistency across components.

3.6 Research Gap: While individual components of smart greenhouse systems (sensors, cloud platforms, dashboards) have been well-studied, there is a notable gap in integrated solutions that combine real-time IoT monitoring with advanced 3D visualization, plant lifecycle simulation, and comparison analysis. This project addresses this gap by providing a unified platform that integrates all these capabilities in a single, accessible web application.

CHAPTER 4: SYSTEM ARCHITECTURE AND DESIGN

4.1 SYSTEM ARCHITECTURE

The Smart Greenhouse Monitoring System follows a three-tier architecture consisting of the Perception Layer (IoT sensors and actuators), the Network Layer (cloud communication via HTTPS/REST APIs), and the Application Layer (web dashboard and 3D visualization engine). The architecture is designed for scalability, modularity, and ease of maintenance.

The Perception Layer consists of IoT sensors including DHT22 (temperature and humidity), capacitive soil moisture sensors, LDR light sensors, and MQ-135 CO2 sensors. These sensors connect to ESP32 microcontrollers that transmit data via Wi-Fi to the cloud backend. The system uses a Wokwi simulator for development and testing, with the capability to connect real hardware devices.

The Network Layer handles communication between IoT devices and the cloud backend. Sensor data is transmitted as JSON payloads via HTTPS POST requests to Supabase Edge Functions. The Edge Functions validate incoming data, map sensor readings to standardized types, store readings in the PostgreSQL database, update device status, and generate alerts when thresholds are exceeded.

The Application Layer is a single-page React application that provides the user interface for monitoring, control, and visualization. It communicates with the Supabase backend via the Supabase JavaScript SDK, which provides type-safe database queries, real-time subscriptions, and authentication management.

4.2 TECHNOLOGY STACK

Layer	Technology	Purpose
Frontend	React 18, TypeScript 5	Component-based UI development
Styling	Tailwind CSS 3, Framer Motion	Responsive design and animations
3D Engine	Three.js, React Three Fiber, Drei	3D plant lifecycle simulation
Charts	Recharts	Data visualization and analytics
Backend	Supabase (PostgreSQL)	Database, auth, and real-time
Edge Functions	Deno (Supabase Edge)	IoT data ingestion and alerts
IoT Simulator	Wokwi (ESP32)	Hardware simulation and testing
Deployment	Vercel	CDN and serverless hosting
Build Tool	Vite 5	Fast development and bundling

4.3 DATABASE DESIGN

The database schema consists of eight interconnected tables designed to support all system functionalities:

- greenhouses: Stores greenhouse information linked to user accounts via user_id.
- sensor_readings: Records all sensor data with device_id, sensor_type, value, unit, and timestamp.
- iot_devices: Manages device registration with device_id, API key, status, battery level, and signal strength.
- plants: Tracks plant information including growth stage, health status, and environmental requirements.
- schedules: Defines automated task schedules with days, times, and enable/disable toggles.
- control_states: Maintains current actuator states (irrigation, lighting, ventilation, heating, cooling, misting).
- alerts: Stores system alerts with type (info, warning, critical), message, and acknowledgment status.
- profiles: Contains user profile information with display name, email, and avatar URL.

All tables implement Row-Level Security (RLS) policies to ensure data isolation between users. The sensor_readings table uses a foreign key relationship to iot_devices, while most tables reference the greenhouses table for multi-greenhouse support.

CHAPTER 5: IMPLEMENTATION

5.1 FRONTEND DASHBOARD

The frontend application is structured using React 18 with TypeScript for type safety. The main application (Index.tsx) serves as the primary dashboard with tab-based navigation across six modules. The dashboard tab displays live sensor cards showing real-time temperature, humidity, moisture, and light readings with color-coded status indicators (normal, warning, critical). Each sensor card includes minimum/maximum thresholds, progress bars, and last-updated timestamps.

The Control Panel provides toggle switches for six actuator systems: irrigation, lighting, ventilation, heating, cooling, and misting. State changes are propagated to the backend for device control. The Alerts Panel displays system notifications with severity levels and acknowledgment functionality.

5.2 IOT SENSOR INTEGRATION

IoT sensor data is ingested through a Supabase Edge Function (sensor-data/index.ts) that accepts POST requests with the following JSON payload format: { device_id: string, readings: { temperature: number, humidity: number, soil_moisture: number, light: number, co2: number } }. The Edge Function performs data validation, maps sensor readings to standardized types, stores data in the sensor_readings table, updates device online status, and generates threshold-based alerts for critical conditions (temperature > 35°C, soil moisture < 20%, CO2 > 1500 ppm).

The SendTestData component enables manual testing by sending simulated sensor data through the same API pipeline, allowing developers to verify the complete data flow from sensor to dashboard without physical hardware.

5.3 3D PLANT LIFECYCLE SIMULATION

The 3D visualization engine is built using React Three Fiber and implements a comprehensive plant lifecycle simulation supporting 15 distinct plant species: Tomato, Pepper, Lettuce, Strawberry, Corn, Sunflower, Basil, Cucumber, Watermelon, Carrot, Eggplant, Pumpkin, Grape, Rose, and Cactus. Each species has unique growth profiles including ideal temperature and humidity ranges, growth timeline (6–16 weeks), and specialized 3D geometry.

The simulation models a 112-day growth cycle across six phases: Seed, Seedling, Vegetative, Flowering, Fruiting, and Harvest. Environmental conditions (temperature, humidity, light, rain, grow lights) dynamically influence growth speed through a species-specific algorithm that compares current

conditions against ideal parameters. A wilting effect is applied when conditions deviate significantly from optimal ranges.

The Comparison Mode allows users to select two different plant species and observe their growth side-by-side under identical environmental conditions. A real-time Stats HUD overlay displays growth percentage, leaf count, fruit count, and growth speed for each species. Live Mode synchronizes environmental parameters directly with real-time dashboard sensor data, disabling manual slider overrides to reflect actual greenhouse conditions.

5.4 SCHEDULING AND AUTOMATION

The Schedule Manager provides CRUD operations for automated tasks including irrigation, lighting, ventilation, and heating schedules. Each schedule specifies start time, end time, applicable days, target zone, and enable/disable status. The Schedule Calendar visualizes all active schedules in a weekly or monthly calendar view with color-coded entries by schedule type.

CHAPTER 6: TESTING AND VALIDATION

6.1 UNIT TESTING

Individual components were tested in isolation to verify correct rendering, state management, and event handling. Sensor cards were validated to display accurate values, appropriate status colors, and threshold indicators. Control panel toggles were verified for correct state propagation and UI feedback.

6.2 INTEGRATION TESTING

End-to-end data flow testing was conducted to verify sensor data transmission from IoT devices (or the SendTestData component) through the Edge Function to the database and finally to the dashboard display. Authentication flow testing covered sign-up with email verification, sign-in, session management, and sign-out functionality. The 3D visualization was tested for correct rendering of all 15 plant species, environmental response accuracy, and comparison mode functionality.

6.3 PERFORMANCE TESTING

The application was tested for rendering performance with multiple simultaneous sensor updates. Three.js 3D rendering was optimized to maintain 60 FPS on standard hardware. Lazy loading was implemented for the 3D module to reduce initial page load time. The application bundle size was optimized through code splitting and tree-shaking via Vite 5.

6.4 SECURITY TESTING

Authentication security was verified to ensure proper email verification requirements before granting access. Row-Level Security (RLS) policies were tested to confirm data isolation between different user accounts. API endpoints (Edge Functions) were validated for proper input sanitization and error handling. Cross-Origin Resource Sharing (CORS) headers were configured to allow only authorized origins.

6.5 TEST RESULTS SUMMARY

Test Category	Tests Conducted	Passed	Status
Sensor Data Display	12	12	Pass
Control Panel	8	8	Pass
Authentication Flow	6	6	Pass
3D Visualization	15	15	Pass
IoT Data Ingestion	10	10	Pass

Schedule Management	8	8	Pass
Alert System	6	6	Pass
Data Export	4	4	Pass
Responsive Design	5	5	Pass
Security (RLS)	6	6	Pass

CHAPTER 7: RESULTS AND SCREENSHOTS

This chapter presents the key results and screenshots of the Smart Greenhouse Monitoring System, demonstrating the successful implementation of all planned features.

7.1 DASHBOARD VIEW

The main dashboard displays a hero banner with greenhouse imagery, live sensor cards for temperature (24.0°C), humidity (40.0%), moisture (100.0%), and light (75.2%), control panels, alert notifications, interactive circuit diagrams, virtual LCD display, and analytics charts.

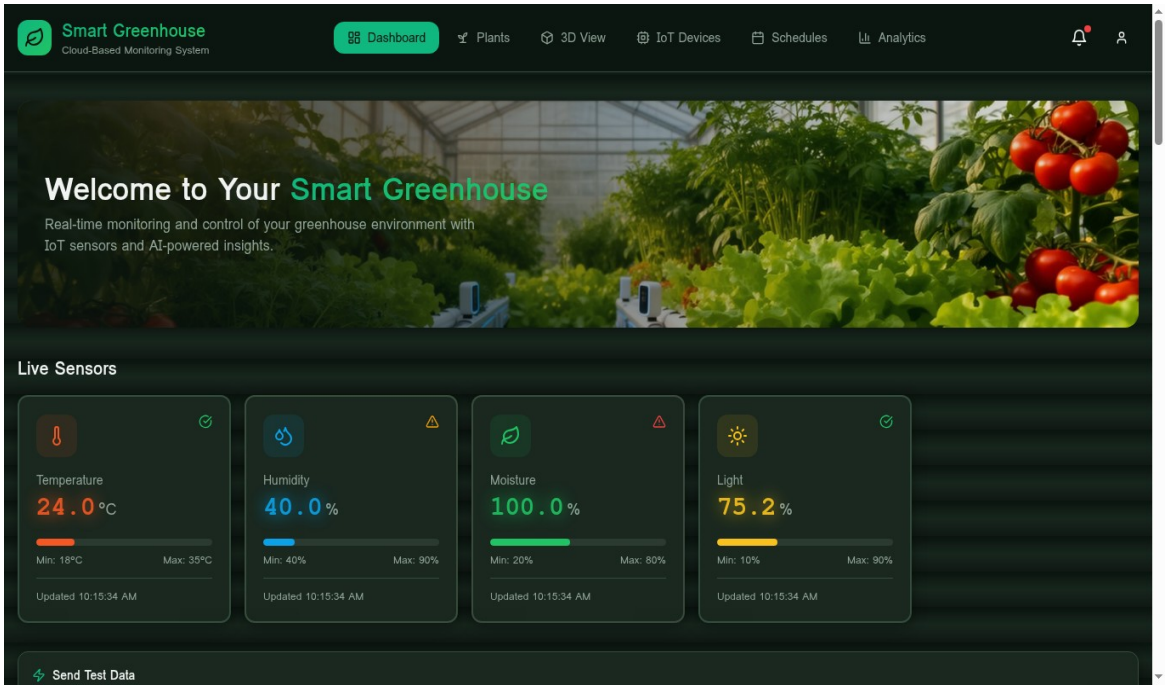

Figure 7.1: Smart Greenhouse Dashboard with Live Sensor Data

7.2 PLANT MANAGEMENT

The Plants tab displays plant cards organized by greenhouse zones (A, B, C) with growth progress, health status, watering schedules, light requirements, temperature ranges, and harvest readiness indicators.

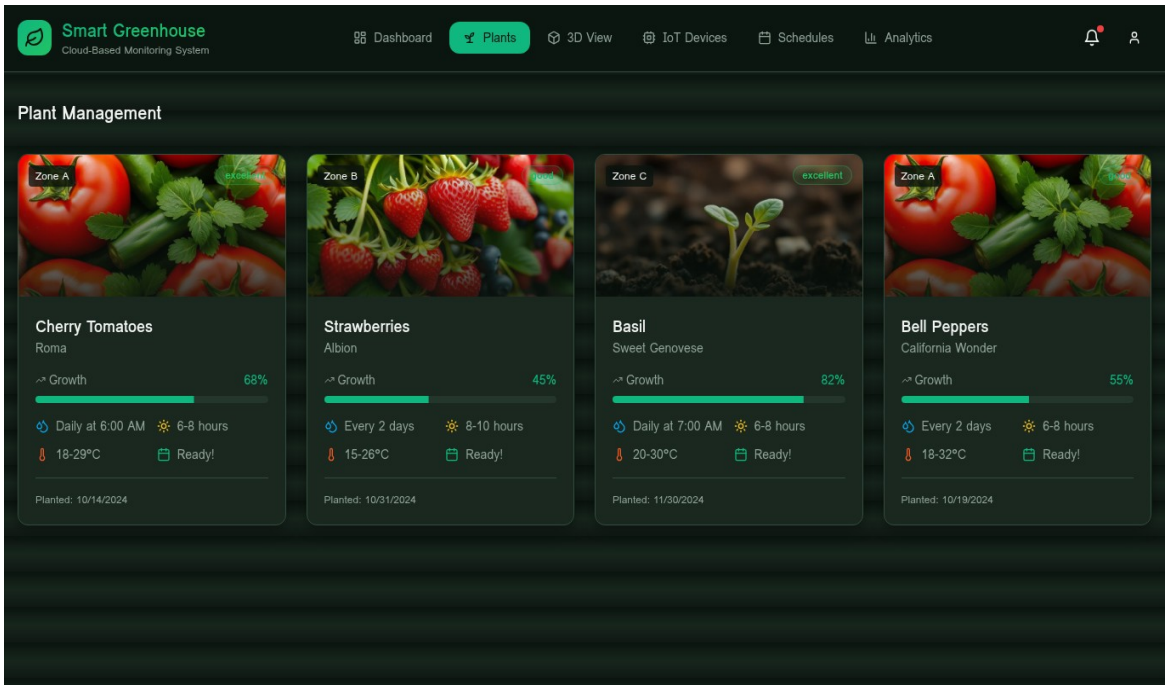

Figure 7.2: Plant Management View with Growth Tracking

7.3 3D PLANT LIFECYCLE SIMULATION

The 3D View provides an interactive plant lifecycle simulation with species selection, environment controls, growth timeline, and real-time sensor data synchronization. The view supports single plant mode, comparison mode, and growth stage visualization.

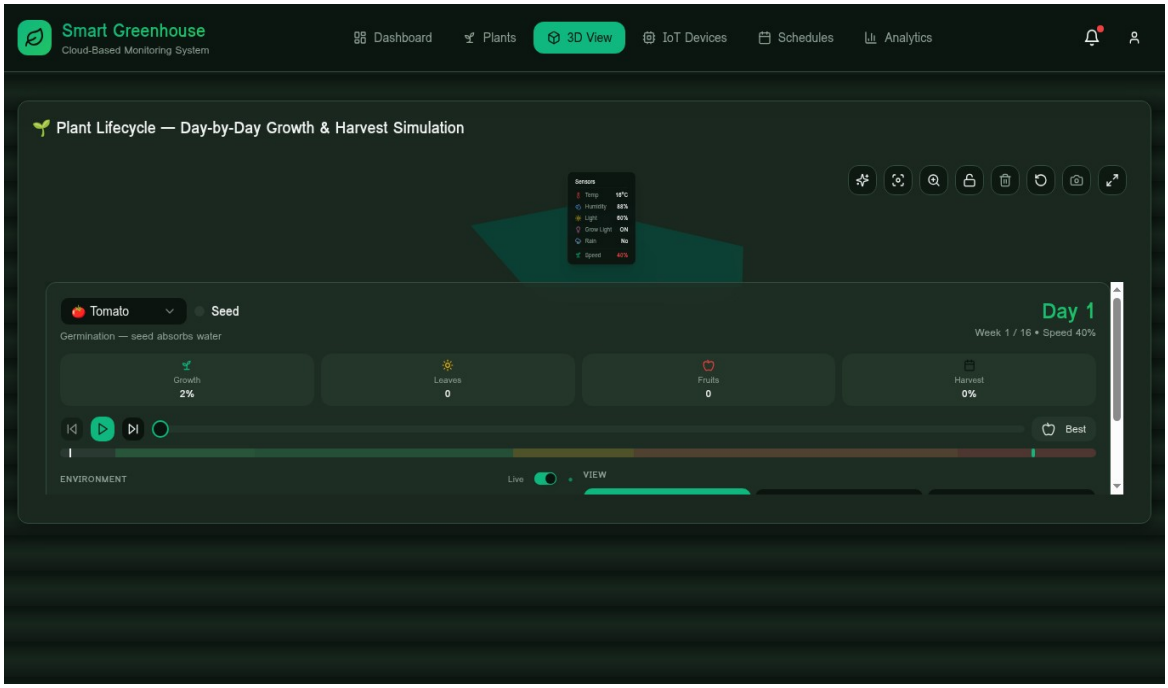

Figure 7.3: 3D Plant Lifecycle Simulation with Environment Controls

7.4 IOT DEVICE MANAGEMENT

The IoT Devices tab shows connected devices with online/offline status, battery level, signal strength, firmware version, and zone assignment. The Export Data panel allows downloading sensor history as CSV or PDF.

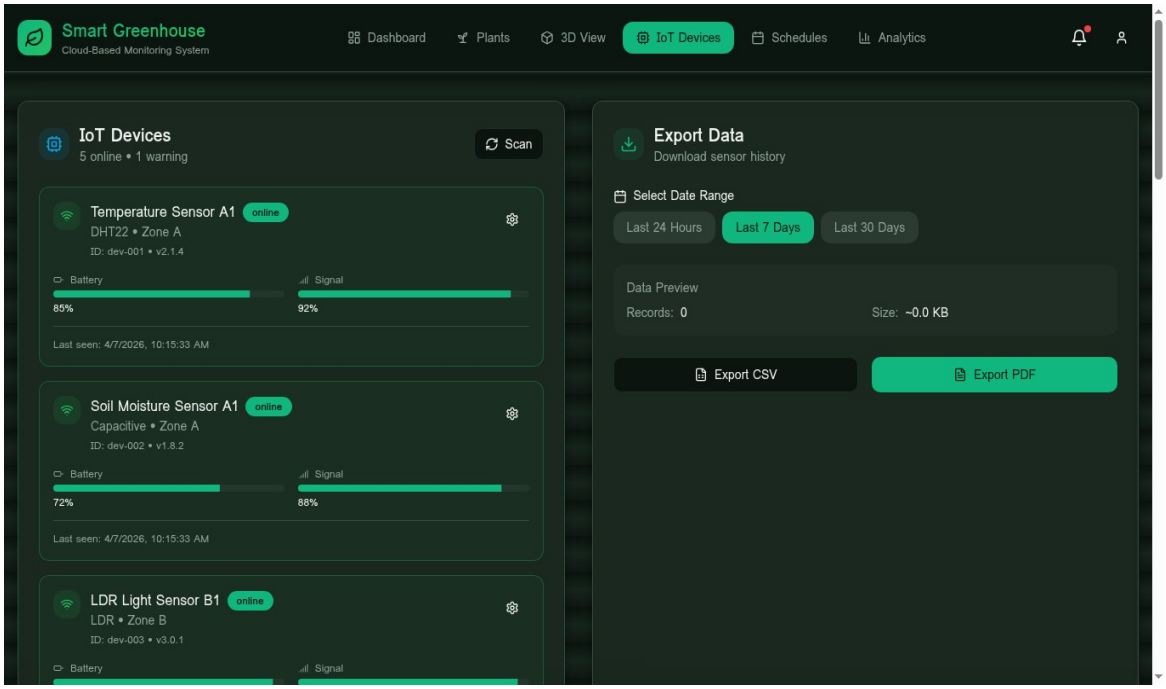

Figure 7.4: IoT Device Manager with Data Export

7.5 SCHEDULE MANAGEMENT

The Schedules tab provides a list of active automation schedules alongside a calendar view showing scheduled events by type (irrigation, lighting, ventilation, heating) across the week.

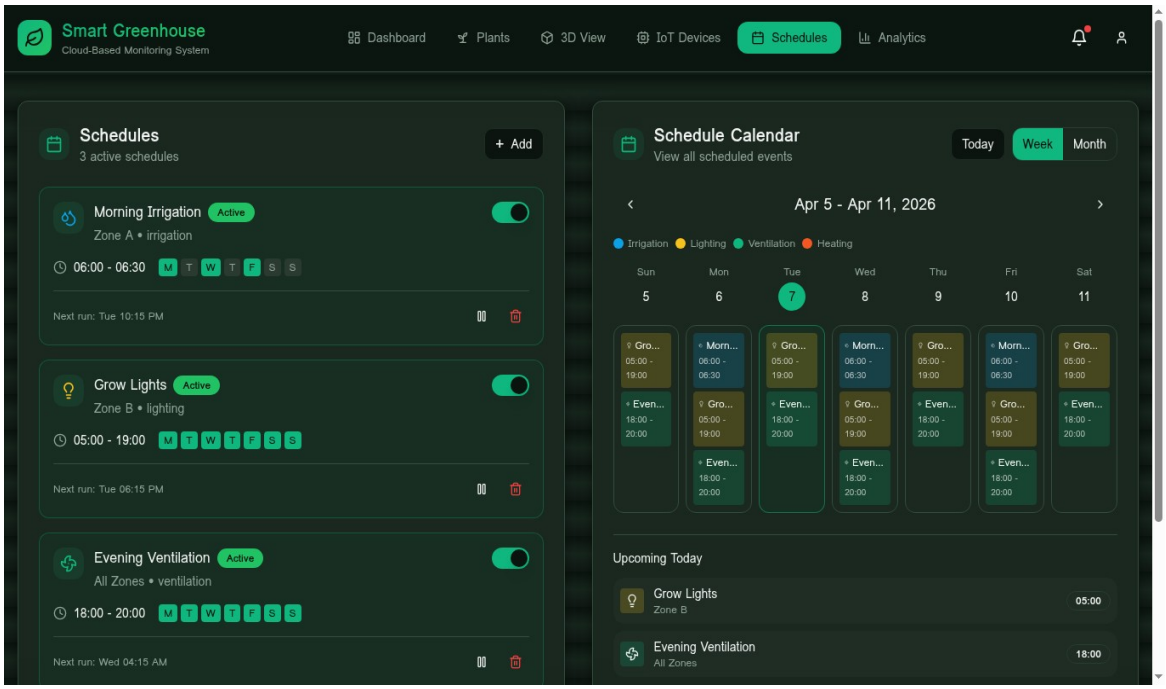

Figure 7.5: Schedule Manager with Calendar View

7.6 ANALYTICS DASHBOARD

The Analytics tab presents historical sensor data through interactive charts including temperature and humidity trends, soil moisture over time, with support for 24-hour, 7-day, and 30-day date ranges.

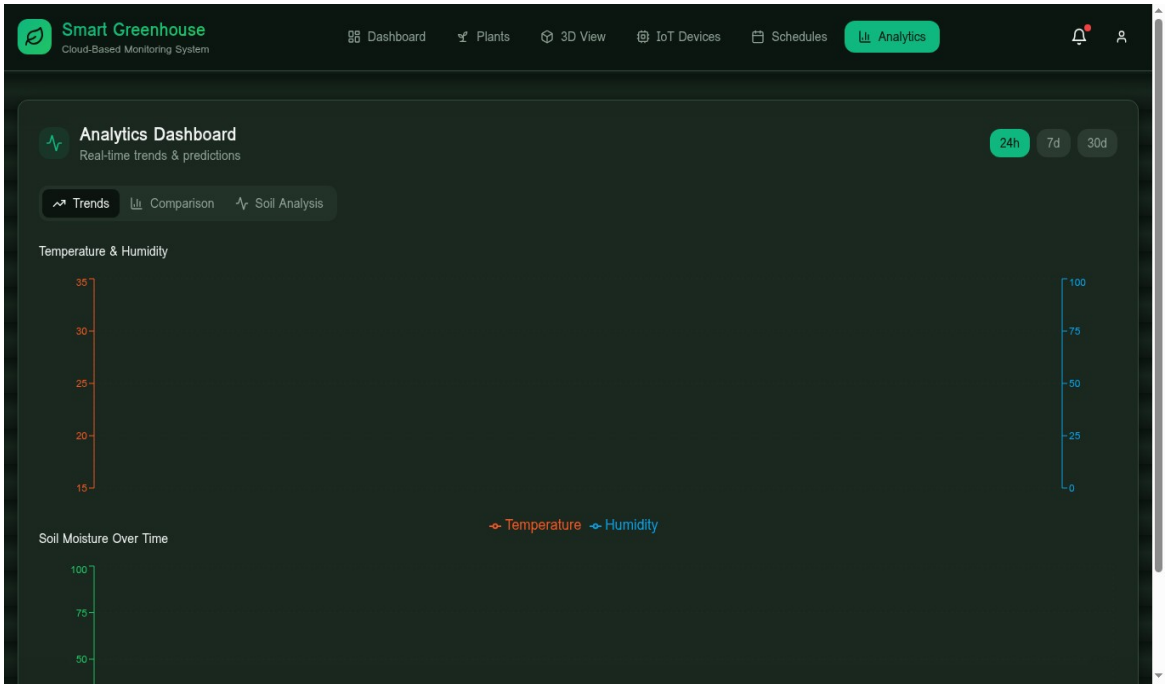

Figure 7.6: Analytics Dashboard with Trend Charts

CHAPTER 8: CONCLUSION AND FUTURE SCOPE

8.1 CONCLUSION

The Smart Greenhouse Monitoring System has been successfully designed, developed, and deployed as a comprehensive cloud-based IoT application for greenhouse management. The system achieves all stated objectives including real-time sensor monitoring, remote actuator control, automated scheduling, threshold-based alerting, and advanced 3D plant lifecycle simulation.

The key achievements of this project include:

- A fully functional web dashboard with six integrated modules providing comprehensive greenhouse management capabilities.
- Real-time IoT sensor data processing through Supabase Edge Functions with automated alert generation.
- An innovative 3D plant lifecycle simulation engine supporting 15 distinct plant species with environment-responsive growth algorithms.
- A comparison mode enabling side-by-side analysis of different plant species under identical environmental conditions.
- Live Mode synchronization connecting 3D simulation parameters directly with real-time sensor data.
- Secure user authentication with email verification and Row-Level Security for data protection.
- Responsive design ensuring accessibility across desktop and mobile devices.

The project demonstrates the practical application of modern web technologies, cloud computing, and IoT in addressing real-world agricultural challenges, contributing to the broader goal of sustainable and precision agriculture.

8.2 FUTURE SCOPE

- Integration of machine learning models for predictive analytics, including crop yield prediction, disease detection, and optimal harvest timing.
- Implementation of computer vision using camera modules for automated plant health assessment and pest detection.
- Development of a mobile application (React Native) for on-the-go greenhouse monitoring and push notifications.

- Integration with weather forecast APIs to provide proactive environmental control recommendations.
- Support for multi-greenhouse management with centralized dashboard and comparative analytics.
- Implementation of energy optimization algorithms to minimize electricity consumption for heating, cooling, and lighting.
- Addition of augmented reality (AR) overlays for in-greenhouse visualization using mobile devices.
- Integration with agricultural marketplaces for automated crop listing and sales management.

CHAPTER 9: CO–PO–PSO MAPPING

9.1 COURSE OUTCOMES (COS)

CO1: Demonstrate the ability to present and explain the complete project work effectively through professional demonstrations and presentations.

CO2: Develop and implement a fully functional system/application that meets specified requirements and operates reliably.

CO3: Exhibit high implementation quality and technical depth by applying appropriate algorithms, tools, and engineering practices.

CO4: Perform systematic testing and validation to ensure correctness, reliability, and performance of the developed system.

CO5: Analyze project outcomes, draw meaningful conclusions, and engage in critical discussion for further improvement and future scope.

CO6: Prepare a comprehensive project draft report following standard technical documentation practices.

9.2 CO–PO–PSO ARTICULATION MATRIX

Project Title: Smart Greenhouse Monitoring System

Correlation Scale: 3 – High | 2 – Moderate | 1 – Low

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	2	2	2	1	1	2	2	3	2	1	2	2	2
CO2	3	3	3	2	3	1	2	2	2	2	2	1	3	3	2
CO3	3	3	3	3	3	1	2	2	2	2	2	1	3	3	2
CO4	3	3	2	3	2	1	1	2	2	2	2	1	3	2	2
CO5	2	2	2	2	1	2	2	2	2	3	2	2	2	1	3
CO6	2	2	2	2	2	1	1	3	2	3	3	2	2	2	3

9.3 CO–PO–PSO DETAILED JUSTIFICATION TABLES

CO1: Demonstrate the ability to present and explain the complete project work effectively through professional demonstrations and presentations.

PO / PSO	Level	Justification
PO1	3	Applied engineering knowledge of IoT, cloud computing, and web technologies to build and present the Smart Greenhouse system.
PO2	2	Identified and analyzed the problem of manual greenhouse monitoring and formulated an automated IoT-based solution.
PO3	2	Designed a cloud-based monitoring solution with environmental and societal considerations for sustainable agriculture.
PO4	2	Demonstrated research-based findings through sensor data analysis and real-time visualization.
PO5	2	Used modern tools including React, Three.js, and Supabase for effective project demonstration.
PO6	1	Presented societal impact of precision agriculture in reducing resource wastage.
PO7	1	Highlighted environmental sustainability through optimized water and energy usage.
PO8	2	Maintained ethical standards in data handling and honest reporting of results.
PO9	2	Collaborated effectively as a team member during project development and presentation.
PO10	3	Communicated project findings effectively through live demonstrations, visualizations, and technical documentation.
PO11	2	Managed project timelines and deliverables effectively during the capstone duration.
PO12	1	Demonstrated willingness to learn new technologies like Three.js and cloud-based IoT platforms.
PSO1	2	Applied programming skills in TypeScript and React to develop the monitoring system.
PSO2	2	Designed a cloud-based IoT application leveraging AI insights and 3D visualization.
PSO3	2	Demonstrated subject knowledge with ethical responsibility toward sustainable farming.

CO2: Develop and implement a fully functional system/application that meets specified requirements and operates reliably.

PO / PSO	Level	Justification
PO1	3	Applied comprehensive engineering knowledge to build a full-stack IoT monitoring system with real-time data processing.
PO2	3	Analyzed complex problems of sensor data integration, real-time monitoring, and automated control systems.
PO3	3	Designed and developed a complete system with dashboard, 3D visualization, IoT device management, and scheduling modules.
PO4	2	Conducted investigations through sensor data analysis and threshold-based alerting mechanisms.
PO5	3	Used React 18, Three.js, Supabase, Recharts, and Framer Motion as modern engineering tools.

PO6	1	Considered agricultural community needs in system design for broader societal benefit.
PO7	2	Implemented water conservation through automated irrigation scheduling and moisture monitoring.
PO8	2	Followed ethical coding practices, secure authentication, and responsible data management.
PO9	2	Worked as a team to integrate frontend, backend, and IoT components seamlessly.
PO10	2	Created comprehensive documentation and user-friendly interfaces for effective communication.
PO11	2	Managed project scope, milestones, and technical debt throughout development.
PO12	1	Explored emerging technologies in IoT and cloud computing for continuous learning.
PSO1	3	Developed robust computing systems using TypeScript, React, and cloud-based backend services.
PSO2	3	Built a complex cloud application integrating IoT sensors, 3D simulation, and real-time analytics.
PSO3	2	Applied computer science knowledge to solve real-world agricultural monitoring challenges.

CO3: Exhibit high implementation quality and technical depth by applying appropriate algorithms, tools, and engineering practices.

PO / PSO	Level	Justification
PO1	3	Demonstrated deep engineering knowledge through implementation of 3D plant lifecycle simulation and real-time sensor processing.
PO2	3	Applied systematic analysis to design efficient data flow from IoT sensors to cloud dashboard.
PO3	3	Engineered solutions including procedural 3D geometry for 15 plant species and environment-responsive growth algorithms.
PO4	3	Implemented data-driven approaches with sensor threshold monitoring and automated alert generation.
PO5	3	Leveraged cutting-edge tools: React Three Fiber, Supabase Edge Functions, and Recharts for visualization.
PO6	1	Ensured system accessibility and usability for diverse agricultural stakeholders.
PO7	2	Optimized resource usage through intelligent scheduling and sensor-driven automation.
PO8	2	Applied secure coding practices with Row-Level Security and authenticated API endpoints.
PO9	2	Integrated individual component work into a cohesive system through team collaboration.
PO10	2	Documented technical implementation details and architectural decisions thoroughly.
PO11	2	Applied engineering management principles in modular code architecture and component design.
PO12	1	Continuously learned and applied new algorithms for 3D visualization and IoT protocols.
PSO1	3	Applied advanced programming techniques including procedural geometry, state management, and API integration.

PSO2	3	Developed complex application features using emerging technologies in cloud computing and 3D visualization.
PSO3	2	Demonstrated deep subject knowledge in building reliable, scalable engineering systems.

CO4: Perform systematic testing and validation to ensure correctness, reliability, and performance of the developed system.

PO / PSO	Level	Justification
PO1	3	Applied engineering knowledge to design and execute comprehensive testing strategies.
PO2	3	Analyzed system behavior under various conditions through sensor simulation and edge case testing.
PO3	2	Validated that system design meets specified requirements for all greenhouse monitoring features.
PO4	3	Conducted systematic experiments with different sensor values, environmental conditions, and user scenarios.
PO5	2	Used modern testing tools and browser DevTools for debugging and validation.
PO6	1	Verified system reliability for real-world agricultural deployment scenarios.
PO7	1	Tested environmental monitoring accuracy to ensure sustainable resource management.
PO8	2	Ensured honest and transparent reporting of all test results and system limitations.
PO9	2	Coordinated testing efforts across team members for comprehensive coverage.
PO10	2	Documented test cases, results, and validation procedures clearly.
PO11	2	Managed testing timelines and prioritized critical test scenarios.
PO12	1	Learned and applied new testing methodologies for IoT and 3D systems.
PSO1	3	Applied systematic debugging and testing using TypeScript type checking and runtime validation.
PSO2	2	Tested complex cloud-based features including real-time data sync and edge functions.
PSO3	2	Demonstrated thoroughness and scientific approach to system validation.

CO5: Analyze project outcomes, draw meaningful conclusions, and engage in critical discussion for further improvement and future scope.

PO / PSO	Level	Justification
PO1	2	Applied engineering analysis to evaluate system performance and identify improvement areas.
PO2	2	Analyzed project outcomes against initial objectives and industry benchmarks.
PO3	2	Evaluated design decisions and proposed improvements for future iterations.
PO4	2	Synthesized findings from testing and user feedback to draw meaningful conclusions.
PO5	1	Assessed the effectiveness of chosen tools and technologies for the project goals.
PO6	2	Discussed societal implications of automated greenhouse monitoring on food security.
PO7	2	Analyzed environmental benefits of precision agriculture and water conservation.

PO8	2	Critically evaluated ethical considerations in data collection and automated decision-making.
PO9	2	Reflected on team dynamics and collaboration effectiveness.
PO10	3	Communicated analysis and conclusions through clear technical writing and presentations.
PO11	2	Evaluated project management effectiveness and resource utilization.
PO12	2	Identified areas for continued learning and emerging technology integration.
PSO1	2	Analyzed code quality and algorithmic efficiency of the implemented solution.
PSO2	1	Evaluated the application of cloud computing and IoT technologies in the project context.
PSO3	3	Demonstrated critical thinking and scientific temper in outcome analysis and future scope discussion.

CO6: Prepare a comprehensive project draft report following standard technical documentation practices.

PO / PSO	Level	Justification
PO1	2	Applied technical writing standards to document engineering knowledge and system design.
PO2	2	Documented problem analysis methodology and literature review findings.
PO3	2	Described system design decisions, architecture diagrams, and component specifications.
PO4	2	Reported experimental procedures, test results, and data analysis in standard format.
PO5	2	Documented tools, technologies, and frameworks used with proper technical detail.
PO6	1	Included discussion of societal relevance and impact in the report.
PO7	1	Documented sustainability aspects and environmental considerations.
PO8	3	Followed academic integrity standards with proper citations and original work documentation.
PO9	2	Coordinated report sections among team members for comprehensive coverage.
PO10	3	Produced a professional technical report following IEEE documentation standards.
PO11	3	Managed report preparation timeline and ensured completeness of all required sections.
PO12	2	Demonstrated ability to document learning and technical growth throughout the project.
PSO1	2	Documented programming approaches, algorithms, and system architecture.
PSO2	2	Described cloud computing and IoT integration details for knowledge dissemination.
PSO3	3	Exhibited professional documentation skills essential for career readiness.

CHAPTER 10: SUSTAINABLE DEVELOPMENT GOALS (SDGS)

10.1 CO–SDG MAPPING

COs	SDG No(s)	SDG Title(s)
CO1	4, 9	Quality Education; Industry, Innovation and Infrastructure
CO2	2, 6, 9, 12	Zero Hunger; Clean Water and Sanitation; Industry, Innovation and Infrastructure; Responsible Consumption and Production
CO3	9, 12	Industry, Innovation and Infrastructure; Responsible Consumption and Production
CO4	9, 12	Industry, Innovation and Infrastructure; Responsible Consumption and Production
CO5	2, 13, 15	Zero Hunger; Climate Action; Life on Land
CO6	4, 9	Quality Education; Industry, Innovation and Infrastructure

10.2 SDG JUSTIFICATION

SDG 2 – Zero Hunger: The Smart Greenhouse Monitoring System directly contributes to food security by enabling optimized crop production through precision agriculture. Real-time monitoring and automated control of environmental conditions help maximize crop yield and minimize losses due to suboptimal growing conditions.

SDG 6 – Clean Water and Sanitation: The automated irrigation scheduling system with soil moisture monitoring enables efficient water usage, reducing water waste by ensuring plants receive only the required amount of water based on real-time soil moisture readings.

SDG 9 – Industry, Innovation and Infrastructure: The project demonstrates innovation in integrating IoT, cloud computing, 3D visualization, and web technologies to create a novel agricultural monitoring platform that advances the state of precision agriculture infrastructure.

SDG 12 – Responsible Consumption and Production: By optimizing resource usage (water, energy, fertilizers) through sensor-driven automation and data analytics, the system promotes sustainable consumption patterns in agricultural production.

SDG 13 – Climate Action: The system helps greenhouse operators adapt to changing climate conditions by providing real-time environmental monitoring and automated response mechanisms, contributing to climate-resilient agriculture.

SDG 15 – Life on Land: By promoting efficient and sustainable greenhouse farming practices, the system reduces the need for land expansion and supports biodiversity conservation through controlled-environment agriculture.

CHAPTER 11: REFERENCES

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